

An investigation of the Quadratic Forms: Basic Properties, the Hurwitz Theorem, the Cubic Norm Form, and Statistical Applications

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Abstract

This paper introduces the geometry and linear–algebraic structure of quadratic forms, focusing on their symmetric matrix representation, orthogonal diagonalization, definiteness, and role in optimization. We then reinterpret classical identities of Diophantus, Brahmagupta, Euler, and Degen as compositions of quadratic forms arising from bilinear maps, leading naturally to the Hurwitz–Radon and Hurwitz theorems and the special dimensions 1, 2, 4, 8, adding a brief extension to a cubic norm form illustrates how similar structure-preserving patterns appear beyond the quadratic case, and an introduction to its application in statistics.

1 Introduction to Quadratic Forms

Quadratic forms appear in many areas of mathematics, often long before we learn to call them by name. What begins in single-variable calculus as the familiar ax^2 term grows in richness and complexity once we move into multiple dimensions. At that point, the algebraic expression stops being just a polynomial and becomes a geometric object, a tool for optimization, and even a window into deeper linear-algebraic structure.

In this section, we aim to give a clear and motivating overview of quadratic forms. Our goal is to keep the explanation mathematically rigorous while also communicating the intuition that has helped me understand and appreciate these objects and theorems more deeply.

1.1 Definition and Matrix Representation

Definition 1.1. A quadratic form on $\mathbb{R}^n$ is a function $Q : \mathbb{R}^n \rightarrow \mathbb{R}, \mathbf{x} \mapsto Q(\mathbf{x})$.

$$Q(\mathbf{x}) = \sum_{i \leq j} c_{ij} x_i x_j$$

This generalizes the second-degree expression $ax^2 + bxy + cy^2$. [Dra26]

Proposition 1.2. *Let $Q : \mathbb{R}^n \rightarrow \mathbb{R}$ be a quadratic form. It can be expressed uniquely as*

$$Q(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$$

Because of the symmetry $x_i x_j = x_j x_i$, the matrix will be $a_{ii} = c_{ii}, a_{ij} = a_{ji} = \frac{1}{2} c_{ij}$. every quadratic form admits a unique symmetric matrix representation:

$$Q(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}, \quad A = A^T,$$

Moreover, because symmetric matrices are determined by their upper triangular entries, the space of all quadratic forms has dimension

$$\dim(Q_n) = \frac{(n+1)n}{2}.$$

1.2 Diagonalization via the Spectral Theorem

The true power of quadratic forms emerges through the Spectral Theorem, which provides a profound simplification. Any real symmetric matrix A can be diagonalized by an orthogonal change of basis. That is, there exists an orthonormal set of eigenvectors

Theorem 1.3 (Spectral Theorem). *If A is a real and symmetric $n \times n$ matrix, then:*

- all eigenvalues of A are real;
- there exists an orthonormal basis $\{\bar{v}_1, \dots, \bar{v}_n\}$ such that $A\bar{v}_i = \lambda_i \bar{v}_i$;
- $A = SDS^{-1} = SDS^T$;

This theorem explains the internal geometry of every quadratic form. We can first express any quadratic form uniquely as:

$$Q(\mathbf{x}) = \mathbf{x}^T A \mathbf{x} = \mathbf{x}^T \lambda \mathbf{x}$$

Then by Spectral Theorem, $\exists B = \{\bar{v}_1, \bar{v}_2 \dots \bar{v}_n\}$ is a orthonormal basis, such that that $A\bar{v}_i = \lambda \bar{v}_i$.

So we can express the quadratic form as:

$$Q(\mathbf{x}) = \lambda_1 c_1^2 + \lambda_2 c_2^2 + \dots + \lambda_n c_n^2$$

All cross terms vanish, and the quadratic form become a pure sum of weighted squares. This expression reveals the geometric structure of the matrix: each eigen-direction is an axis along which the quadratic form changes independently of others, and each eigenvalue measures curvature along that direction. The diagonalization available through Spectral Theorem makes the behavior, geometry, and classification of the quadratic form much clearer.

1.3 Definiteness of Quadratic Forms

1.3.1 Definiteness via Eigenvalues

Once the quadratic form is diagonalized as:

$$Q(\mathbf{x}) = \lambda_1 c_1^2 + \lambda_2 c_2^2 + \dots + \lambda_n c_n^2$$

we can classify the definiteness of the matrix.

- Q is **positive definite** iff $\lambda_i > 0$ for all i .
- Q is **negative definite** iff $\lambda_i < 0$ for all i .
- Q is **indefinite** iff the eigenvalues have mixed signs.
- Q is **positive semidefinite** if each $\lambda_i \geq 0$ and at least one is zero.
- Q is **negative semidefinite** if each $\lambda_i \leq 0$ and at least one is zero.

Geometrically, the diagonalized quadratic form describes a surface whose principal axes correspond to the eigenvectors and whose principal curvatures are the eigenvalues. The sum of weighted squares provides an uncluttered picture: positivity means the surface curves upward in every direction, negativity means it curves downward in every direction, and a mix of signs indicates a saddle-shaped geometry. The spectral perspective therefore offers a direct and intuitive method for understanding how the quadratic form behaves in space.

1.3.2 Definiteness in Calculus: The Hessian Matrix

This classification becomes particularly meaningful in multivariable calculus, where the second-order behavior of a function is governed by its Hessian matrix. For a twice-differentiable function $f : \mathbb{R}^n \rightarrow \mathbb{R}$, the Hessian at a point x_0 is the matrix of second partial derivatives: [AC22]

$$H(x_0) = \begin{pmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} & \cdots & \frac{\partial^2 f}{\partial x_2 \partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \frac{\partial^2 f}{\partial x_n \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_n^2} \end{pmatrix}.$$

Theorem 1.4 (Clairaut's Theorem). *Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a function whose second partial derivatives are continuous in a neighborhood of a point x_0 . Then the order of differentiation does not matter:*

$$\frac{\partial^2 f}{\partial x_i \partial x_j} = \frac{\partial^2 f}{\partial x_j \partial x_i}$$

for all $i, j \leq n$

By Clairaut's theorem on equality of mixed partial derivatives, the Hessian is automatically symmetric. This symmetry makes Hessian matrices compatible with the theory of quadratic forms: the second-order Taylor expansion at x_0 ,

$$f(x_0 + h) = f(x_0) + \frac{1}{2} h^T H(x_0) h + o(\|h\|^2),$$

where H is the Hessian matrix at x_0 . Since H is symmetric by Clairaut's, the term $h^T H h$, which is a quadratic form, determines the local geometry of the graph of f .

Thus, the definiteness of $H(x_0)$ determines the nature of the critical point:

- positive definite $H \Rightarrow$ strict local minimum;
- negative definite $H \Rightarrow$ strict local maximum;
- indefinite $H \Rightarrow$ saddle point.

The diagonalization of H through its eigenvalues provides a geometric interpretation of curvature in each direction, while determinant-based tests offer practical computational tools. In this way, the study of quadratic forms forms the foundation for second-order optimization and reveals the deep connection between linear algebra and local analysis.

1.4 Geometry of Quadratic Surfaces

Diagonalizing a quadratic form does more than simplify its algebra—it uncovers its entire geometric identity. The following content referenced section 6.2 of Gilbert Strang, *Gilbert’s Linear Algebra and Its Applications*. [Str16]

Once the change of basis transforms from $Q(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$ into the diagonal form

$$Q(\mathbf{x}) = \lambda_1 c_1^2 + \cdots + \lambda_n c_n^2,$$

every geometric feature of the surface becomes visible at a glance. The variables c_i now measure the component along orthogonal eigen-directions, and the eigenvalues λ_i act as curvature coefficients along those axes. In this coordinate system, the level sets

$$Q(\mathbf{x}) = k$$

trace out familiar shapes whose behavior is governed entirely by the signs, magnitudes, and multiplicity of the eigenvalues.

The *signature* of the quadratic form,

$$(\#\text{positive } \lambda_i, \#\text{negative } \lambda_i, \#\text{zero eigenvalues}),$$

classifies the shape completely. This signature is invariant under orthogonal change of basis, so it captures the intrinsic geometry rather than the accident of coordinates.

When every eigenvalue is positive, the quadratic form curves upward in all directions, and the level sets $Q(\mathbf{x}) = k > 0$ are ellipsoids. The size and stretch of the ellipsoid are determined by the magnitudes of the eigenvalues: large eigenvalues stretch the surface more steeply along that axis, making the ellipsoid narrower, while small eigenvalues flatten it. Thus, even within the “positive definite” category, the ratios λ_i/λ_j determine how round or anisotropic the surface appears.

When exactly one eigenvalue is negative (or, symmetrically, when exactly one is positive), we enter the world of saddles and hyperboloids. For instance, in three dimensions with signature $(2, 1)$, the surface $Q(\mathbf{x}) = k$ becomes a hyperboloid of one or two sheets, depending on the sign of k . In two dimensions, the same signature produces the familiar hyperbolic paraboloid—a surface that curves upward in one direction and downward in another. This mixed curvature reflects the presence of both positive and negative eigenvalues.

If some eigenvalues vanish, the geometry becomes *degenerate*. When $\lambda_i = 0$, the quadratic form is completely flat along the i -th eigen-direction: the surface neither curves upward nor downward in that direction. Geometrically, this produces cylinders, cones, and “flat valleys,” which are surfaces that bend in some directions but extend freely in others. The rank of the matrix A therefore measures the number of curved directions, while the nullity measures how many dimensions remain flat.

Cross-sections of $Q(\mathbf{x})$ reveal even more. Fixing all but two coordinates reduces the quadratic form to a conic section whose shape (ellipse, hyperbola, or pair of lines) mirrors the signs of the corresponding eigenvalues. As these slices vary, one can see the global surface organized around its principal axes, with curvature changing only in magnitude, never in direction. Orthogonal diagonalization therefore acts like rotating a geometric object until its natural axes are aligned with the coordinate system.

In this way, the diagonal form does not simply simplify computation—it reveals the true geometry of the quadratic surface. The spectrum of the matrix A becomes a complete guide to its shape: eigenvectors determine directions of curvature, eigenvalues determine the intensity of bending, and the signature encodes whether the surface encloses space, straddles space, or lies partially flat. Quadratic forms thus sit at a remarkable intersection between algebra and geometry, where information from a matrix becomes a precise geometric blueprint.

1.5 Summary

Looking back at this section, what strikes me the most is how something as simple as a quadratic equation can be expressed in so many different ways, including a special kind of matrix, and unfold into an entire landscape of geometry. The Quadratic Form feels like a story: a matrix becomes a surface, eigenvalues become curvatures, and the diagonalization becomes the act of rotating my perspective until everything finally lines up and makes sense. As we worked through these ideas, we begin to understand why quadratic forms appear everywhere, from shape of a graph near the critical point to the geometry inside symmetric matrices. This section is only an attempt to capture the basic understanding of Quadratic Forms, and share some intuitions that connected and explained the ideas. In the next section, we will explore more theorems and applications of the Quadratic Form.

2 Composition of Quadratic Form

2.1 Introduction

We begin by presenting an intriguing problem motivated by a series of elegant equations.

Proposition 2.1 (Diophantus identity).

$$(x_1^2 + x_2^2)(y_1^2 + y_2^2) = (x_1y_1 \pm x_2y_2)^2 + (x_1y_2 \mp x_2y_1)^2$$

Proposition 2.2 (Brahmagupta identity).

$$(x_1^2 + nx_2^2)(y_1^2 + ny_2^2) = (x_1y_1 \pm nx_2y_2)^2 + n(x_1y_2 \mp x_2y_1)^2$$

Proposition 2.3 (Euler's four-square identity).

$$\begin{aligned} (x_1^2 + x_2^2 + x_3^2 + x_4^2)(y_1^2 + y_2^2 + y_3^2 + y_4^2) = & (x_1y_1 - x_2y_2 - x_3y_3 - x_4y_4)^2 + \\ & (x_1y_2 + x_2y_1 + x_3y_4 - x_4y_3)^2 + \\ & (x_1y_3 - x_2y_4 + x_3y_1 + x_4y_2)^2 + \\ & (x_1y_4 + x_2y_3 - x_3y_2 + x_4y_1)^2 \end{aligned}$$

Proposition 2.4 (Degen's eight-square identity).

$$\begin{aligned} (x_1^2 + x_2^2 + x_3^2 + x_4^2 + x_5^2 + x_6^2 + x_7^2 + x_8^2)(y_1^2 + y_2^2 + y_3^2 + y_4^2 + y_5^2 + y_6^2 + y_7^2 + y_8^2) = & \\ (x_1y_1 - x_2y_2 - x_3y_3 - x_4y_4 - x_5y_5 - x_6y_6 - x_7y_7 - x_8y_8)^2 + & \\ (x_1y_2 + x_2y_1 + x_3y_4 - x_4y_3 + x_5y_6 - x_6y_5 - x_7y_8 + x_8y_7)^2 + & \\ (x_1y_3 - x_2y_4 + x_3y_1 + x_4y_2 + x_5y_7 + x_6y_8 - x_7y_5 - x_8y_6)^2 + & \\ (x_1y_4 + x_2y_3 - x_3y_2 + x_4y_1 + x_5y_8 - x_6y_7 + x_7y_6 - x_8y_5)^2 + & \\ (x_1y_5 - x_2y_6 - x_3y_7 - x_4y_8 + x_5y_1 + x_6y_2 + x_7y_3 + x_8y_4)^2 + & \\ (x_1y_6 + x_2y_5 - x_3y_8 + x_4y_7 - x_5y_2 + x_6y_1 - x_7y_4 + x_8y_3)^2 + & \\ (x_1y_7 + x_2y_8 + x_3y_5 - x_4y_6 - x_5y_3 + x_6y_4 + x_7y_1 - x_8y_2)^2 + & \\ (x_1y_8 - x_2y_7 + x_3y_6 + x_4y_5 - x_5y_4 - x_6y_3 + x_7y_2 + x_8y_1)^2 & \end{aligned}$$

Notice that the examples above all preserve their algebraic structure under multiplication: the product of two sums of squares can again be expressed as a sum of squares of the same form. These identities arise specifically for sums of 2^k squares. It would be natural to conjecture that analogous identities exist for sixteen squares, thirty-two squares, and so on. However, these are the only cases in which such a structure-preserving identity exists.

This problem is known as the composition of quadratic forms, a classical and still active area of research in mathematics. The study originated in Gauss's work on binary quadratic forms, with later fundamental contributions by Hurwitz and, more recently, Bhargava.

In the sections below, we focus on the Hurwitz Problem, which concerns quadratic forms of the type

$$Q(\mathbf{x}) = \mathbf{x}^T I_n \mathbf{x},$$

Equivalently, if $\mathbf{x} = (x_1, \dots, x_n) \in \mathbb{F}^n$, then

$$Q(\mathbf{x}) = \sum_{i=1}^n x_i^2$$

2.2 Hurwitz Problem

Before stating the Hurwitz problem, we introduce precise definitions for the mathematical notions involved.

Definition 2.5 (Bilinear Map). Let U, V, W be vector spaces over a field $\mathbb{F}$. A map

$$B : U \times V \rightarrow W$$

is called bilinear if it is linear in each component. That is, for all $\bar{u}_1, \bar{u}_2 \in U$, $\bar{v}_1, \bar{v}_2 \in V$, and scalars $\lambda_1, \lambda_2 \in \mathbb{F}$, the following hold:

$$B(\lambda_1 \bar{u}_1 + \lambda_2 \bar{u}_2, \bar{v}) = \lambda_1 B(\bar{u}_1, \bar{v}) + \lambda_2 B(\bar{u}_2, \bar{v}), \quad \forall \bar{v} \in V,$$

$$B(\bar{u}, \lambda_1 \bar{v}_1 + \lambda_2 \bar{v}_2) = \lambda_1 B(\bar{u}, \bar{v}_1) + \lambda_2 B(\bar{u}, \bar{v}_2), \quad \forall \bar{u} \in U.$$

Definition 2.6 (Standard Quadratic Form). For $\mathbf{x} = (x_1, \dots, x_n) \in \mathbb{F}^n$, the Standard Quadratic Form will be

$$Q(\mathbf{x}) = \mathbf{x}^T I_n \mathbf{x} = \sum_{i=1}^n x_i^2$$

Definition 2.7 (Hurwitz Problem). Let $Q(\mathbf{x}) = x_1^2 + \dots + x_n^2$ be the standard quadratic form on $\mathbb{F}^n$. For which positive integers n does there exist a bilinear map $\star : \mathbb{F}^n \times \mathbb{F}^n \rightarrow \mathbb{F}^n$ such that

$$Q(\mathbf{x} \star \mathbf{y}) = Q(\mathbf{x}) Q(\mathbf{y})$$

for all $\mathbf{x}, \mathbf{y} \in \mathbb{F}^n$?

2.3 Hurwitz–Radon Theorem and its Proof

Lemma 2.8 (Bilinear Form Expression). $\mathbf{x}, \mathbf{y} \in \mathbb{F}^n$, and $\star$ is a bilinear map. Let $\mathbf{z} = \mathbf{x} \star \mathbf{y}$, then

$$\mathbf{z} = \sum_{i=1}^n \sum_{j=1}^n a_{ij} x_i y_j$$

Proof. Let $\mathbf{e}_1, \dots, \mathbf{e}_n$ be the standard basis of $\mathbb{F}^n$. Then, $\mathbf{x} = \sum_{i=1}^n x_i \mathbf{e}_i$, $\mathbf{y} = \sum_{j=1}^n y_j \mathbf{e}_j$. By bilinearity,

$$\mathbf{x} \star \mathbf{y} = \sum_{i=1}^n \sum_{j=1}^n (\mathbf{e}_i \star \mathbf{e}_j) x_i y_j$$

Therefore, let $a_{ij} = \mathbf{e}_i \star \mathbf{e}_j$, and the result holds. $\square$

By Lemma 2.8, let the k th entry of $\mathbf{z}$ be $z_k = \sum_{i=1}^n \sum_{j=1}^n a_{ijk} x_i y_j$. Then,

$$\begin{aligned} \begin{pmatrix} z_1 \\ \vdots \\ z_n \end{pmatrix} &= \begin{pmatrix} \sum_{i=1}^n \sum_{j=1}^n a_{ij1} x_i y_j \\ \vdots \\ \sum_{i=1}^n \sum_{j=1}^n a_{ijn} x_i y_j \end{pmatrix} \\ &= \begin{pmatrix} \sum_{j=1}^n (\sum_{i=1}^n a_{ij1} x_i) y_j \\ \vdots \\ \sum_{j=1}^n (\sum_{i=1}^n a_{ijn} x_i) y_j \end{pmatrix} \\ &= \begin{pmatrix} \sum_{i=1}^n a_{i11} x_i & \cdots & \sum_{i=1}^n a_{in1} x_i \\ \vdots & \ddots & \vdots \\ \sum_{i=1}^n a_{i1n} x_i & \cdots & \sum_{i=1}^n a_{inn} x_i \end{pmatrix} \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} \\ &= \begin{pmatrix} x_1 \begin{pmatrix} a_{111} & \cdots & a_{1n1} \\ \vdots & \ddots & \vdots \\ a_{11n} & \cdots & a_{1nn} \end{pmatrix} + \cdots + x_n \begin{pmatrix} a_{n11} & \cdots & a_{nn1} \\ \vdots & \ddots & \vdots \\ a_{n1n} & \cdots & a_{nnn} \end{pmatrix} \end{pmatrix} \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} \end{aligned}$$

Let $A_{\mathbf{x}} = x_1 A_1 + \cdots + x_n A_n$. Then, we can express

$$\mathbf{z} = (x_1 A_1 + \cdots + x_n A_n) \mathbf{y} = A_{\mathbf{x}} \mathbf{y}$$

such that A_i are $n \times n$ matrices with the (j, k) th entry of A_i being a_{ikj} .

One can see this intuitively when $n = 2$ by writing

$$\mathbf{z} = \begin{pmatrix} z_1 \\ z_2 \end{pmatrix} = \begin{pmatrix} x_1 y_1 - x_2 y_2 \\ x_1 y_2 + x_2 y_1 \end{pmatrix} = \begin{pmatrix} x_1 & -x_2 \\ x_2 & x_1 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \left(x_1 \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + x_2 \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \right) \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$$

Now, we evaluate $Q(\mathbf{z})$.

$$\begin{aligned} Q(\mathbf{z}) &= \mathbf{z}^T I_n \mathbf{z} \\ &= \mathbf{y}^T A_{\mathbf{x}}^T A_{\mathbf{x}} \mathbf{y} \\ &= (A_{\mathbf{x}}^T A_{\mathbf{x}} \mathbf{y}) \cdot \mathbf{y} \end{aligned}$$

Because $Q(\mathbf{z}) = Q(\mathbf{x})Q(\mathbf{y})$, hence

$$Q(\mathbf{z}) = \mathbf{x}^T I_n \mathbf{x} \mathbf{y}^T I_n \mathbf{y} = \left(\sum_{i=1}^n x_i^2 \right) \mathbf{y} \cdot \mathbf{y} = \left(\left(\sum_{i=1}^n x_i^2 \right) \mathbf{y} \right) \cdot \mathbf{y}$$

Because this holds for all $\mathbf{y} \in \mathbb{F}^n$, hence

$$A_{\mathbf{x}}^T A_{\mathbf{x}} = \left(\sum_{i=1}^n x_i^2 \right) I_n$$

Recall that $A_{\mathbf{x}} = x_1 A_1 + \cdots + x_n A_n$. Therefore, we have

$$A_{\mathbf{x}}^T A_{\mathbf{x}} = \sum_{i=1}^n (A_i^T A_i) x_i^2 + \sum_{1 \leq i < j \leq n} (A_i^T A_j + A_j^T A_i) x_i x_j$$

Therefore, we must have

$$A_i^T A_i = I_n, A_i^T A_j + A_j^T A_i = O$$

By the first result, we know that all A_i are orthogonal matrices, and their determinant will be ± 1 .

By the second result, we know that $A_i^T A_j = -A_j^T A_i$. Taking the determinant on both sides, we can get $\det(A_i) \det(A_j) = (-1)^n \det(A_j) \det(A_i)$, which implies that n must be even (with the exception of 1 since the argument requires the existence of two distinct indices $i \neq j$).

This is the furthest we can reach by linear algebra. With some group theory and representation theory, we can prove that [Con15]

$$2^{\frac{n}{2}-1} \mid n$$

Therefore, $n \leq 8$, and the only possibilities are $n = 2, 4, 6, 8$. Since $2^2 \nmid 6$, hence when n is even, the only possibilities are $n = 2, 4, 8$, and they correspond to the Diophantus, Euler's four square, and Degen's eight square identities. Now, we have reached the theorem.

Theorem 2.9 (Hurwitz–Radon Theorem). *Let $Q(\mathbf{x}) = x_1^2 + \cdots + x_n^2$ be the standard quadratic form on $\mathbb{F}^n$. There exists a bilinear map $\star : \mathbb{F}^n \times \mathbb{F}^n \rightarrow \mathbb{F}^n$ such that*

$$Q(\mathbf{x} \star \mathbf{y}) = Q(\mathbf{x})Q(\mathbf{y})$$

if and only if $n = 1, 2, 4, 8$.

2.4 Hurwitz Theorem

The version proved above concerns only the standard quadratic form

$$Q(\mathbf{x}) = x_1^2 + \cdots + x_n^2.$$

One may ask for which quadratic forms Q over a field $\mathbb{F}$ there exists a bilinear map $\star : \mathbb{F}^n \times \mathbb{F}^n \rightarrow \mathbb{F}^n$ such that

$$Q(\mathbf{x} \star \mathbf{y}) = Q(\mathbf{x})Q(\mathbf{y}).$$

Assume Q is a nondegenerate positive definite quadratic form. Then Q can be written as

$$Q(\mathbf{x}) = \mathbf{x}^T A \mathbf{x},$$

where A is a symmetric, positive definite matrix. By the spectral theorem 1.3, there exists an invertible matrix S such that

$$A = S^T D S,$$

where D is a diagonal matrix with positive diagonal entries. Hence

$$Q(\mathbf{x}) = (S\mathbf{x})^T D (S\mathbf{x}).$$

Since D has positive diagonal entries, a further linear rescaling of variables reduces Q to the standard quadratic form

$$Q(\mathbf{x}) = x_1^2 + \cdots + x_n^2.$$

Therefore, the existence of a bilinear composition law for Q is invariant under linear changes of variables and reduces to the standard case.

Theorem 2.10 (Hurwitz Theorem). *Let $Q : \mathbb{R}^n \rightarrow \mathbb{R}$ be a positive definite quadratic form. There exists a bilinear map $\star : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}^n$ such that*

$$Q(\mathbf{x} \star \mathbf{y}) = Q(\mathbf{x})Q(\mathbf{y})$$

if and only if $\mathbb{R}^n$ can be equipped with the structure of a normed division algebra over $\mathbb{R}$; equivalently, $n = 1, 2, 4, 8$, corresponding to $\mathbb{R}, \mathbb{C}, \mathbb{H}, \mathbb{O}$.

3 Extension: A Cubic Norm Form

Quadratic forms are special partly because they sit at the boundary between algebra and linear geometry: they can be encoded by symmetric matrices, diagonalized orthogonally, and classified by eigenvalues. It is natural to ask what survives if we move one degree higher. In this extension we highlight a cubic form that behaves *like* a norm: it is preserved under a bilinear-type multiplication, even though it no longer comes from a symmetric matrix or admits an orthogonal diagonalization.

3.1 A cubic form and where it comes from

Definition 3.1 (Cubic form). A *cubic form* on $\mathbb{R}^n$ is a homogeneous polynomial $C : \mathbb{R}^n \rightarrow \mathbb{R}$ of total degree 3, i.e.

$$C(\lambda x) = \lambda^3 C(x) \quad \text{for all } \lambda \in \mathbb{R}, x \in \mathbb{R}^n.$$

The most relevant cubic form for our purposes is the *cubic norm form*

$$N(a, b, c) = a^3 + b^3 + c^3 - 3abc.$$

This expression appears naturally in several places (e.g. symmetric polynomials), but the viewpoint that matches the rest of this paper is the determinantal one.

3.2 Determinantal interpretation

Define the 3×3 matrix

$$M(a, b, c) = \begin{pmatrix} a & b & c \\ c & a & b \\ b & c & a \end{pmatrix}.$$

Proposition 3.2. *For all $a, b, c \in \mathbb{R}$,*

$$\det(M(a, b, c)) = a^3 + b^3 + c^3 - 3abc = N(a, b, c).$$

Proof. Expand $\det(M(a, b, c))$ along the first row:

$$\det(M) = a \det \begin{pmatrix} c & b \\ b & a \end{pmatrix} - b \det \begin{pmatrix} c & a \\ b & c \end{pmatrix} + c \det \begin{pmatrix} a & a \\ b & c \end{pmatrix}.$$

Compute each 2×2 determinant:

$$\det \begin{pmatrix} a & b \\ c & a \end{pmatrix} = a^2 - bc, \quad \det \begin{pmatrix} c & b \\ b & a \end{pmatrix} = ac - b^2, \quad \det \begin{pmatrix} c & a \\ b & c \end{pmatrix} = c^2 - ab.$$

Substitute back:

$$\begin{aligned}\det(M) &= a(a^2 - bc) - b(ac - b^2) + c(c^2 - ab) \\ &= a^3 - abc - abc + b^3 + c^3 - abc \\ &= a^3 + b^3 + c^3 - 3abc,\end{aligned}$$

which is exactly $N(a, b, c)$. $\square$

This is the cubic analogue of the matrix representation $Q(x) = x^T A x$ for quadratic forms, except the matrix here is not encoding N via a bilinear pairing; instead it packages N as a determinant.

3.3 A norm-preserving (composition) structure

The most striking feature of N is that it is *multiplicative* under a natural bilinear-type product on $\mathbb{R}^3$ [Abi95]. Define, for $x = (a, b, c)$ and $y = (d, e, f)$,

$$x \star y = (u, v, w),$$

where

$$u = ad + bf + ce, \quad v = ae + bd + cf, \quad w = af + be + cd. \quad (1)$$

Lemma 3.3. *For all $x = (a, b, c)$ and $y = (d, e, f)$ in $\mathbb{R}^3$,*

$$M(a, b, c) M(d, e, f) = M(u, v, w),$$

where $(u, v, w) = x \star y$ as in (1).

Proof. Compute the first row of the product $M(a, b, c)M(d, e, f)$:

$$(a, b, c) \begin{pmatrix} d & e & f \\ f & d & e \\ e & f & d \end{pmatrix} = (ad + bf + ce, ae + bd + cf, af + be + cd) = (u, v, w).$$

Because the product of two circulant matrices is again circulant, the remaining rows are forced to be the cyclic shifts of (u, v, w) , hence the product equals $M(u, v, w)$. $\square$

Theorem 3.4 (Cubic norm multiplicativity). *For all $x, y \in \mathbb{R}^3$,*

$$N(x \star y) = N(x) N(y).$$

Equivalently,

$$(a^3 + b^3 + c^3 - 3abc)(d^3 + e^3 + f^3 - 3def) = u^3 + v^3 + w^3 - 3uvw,$$

with (u, v, w) given by (1).

Proof. By Lemma 3.3 we have $M(x)M(y) = M(x \star y)$, so taking determinants gives

$$\det(M(x \star y)) = \det(M(x)M(y)) = \det(M(x)) \det(M(y)).$$

By Proposition 3.2, $\det(M(\cdot)) = N(\cdot)$, hence

$$N(x \star y) = N(x) N(y).$$

$\square$

This is the precise sense in which N behaves like a “norm”: the value of the form is preserved under a bilinear-type multiplication $\star$, exactly mirroring how sums of squares are preserved under the bilinear compositions in Section 2.

3.4 Limits of the cubic setting

Although $N(a, b, c)$ shares a composition law with the quadratic norms, the linear-algebraic tools that made quadratic forms so clean do *not* extend in the same way.

- **No symmetric matrix representation.** A quadratic form is determined by a symmetric bilinear pairing and can be written as $Q(x) = x^T A x$. A general cubic form cannot be encoded by a single matrix in this way. Its coefficients naturally assemble into an order-3 tensor, not a 2-tensor (matrix), and the change-of-basis action is correspondingly more complicated.
- **No orthogonal diagonalization.** For quadratic forms, the spectral theorem provides an orthonormal eigenbasis so that cross terms vanish. For cubic forms there is no analogue that uniformly removes mixed terms via an orthogonal change of variables; even defining a useful notion of “eigenvalues” for a general cubic tensor is subtle and depends on additional structure.
- **Classification is harder.** Definiteness and signature give a complete and stable classification for real quadratic forms under orthogonal changes of basis. Cubic forms do not admit such a simple invariant-based classification: two cubic forms can share many naive invariants and still be inequivalent, and the geometry of their level sets can change in more delicate ways.

In short, cubic norm forms can still exhibit the same *structure-preserving* phenomenon that motivated the Hurwitz problem, but the organizing linear-algebraic framework for quadratic forms (symmetric matrices, eigenvalues, orthogonal diagonalization) is genuinely special to degree 2.

4 Application in Statistics

Quadratic forms are not merely abstract algebraic objects - they are also the mathematical backbone of classical statistical inference, where their linear algebraic structure (symmetry, diagonalization, definiteness) and geometric interpretation (projection, squared lengths) become indispensable for quantifying uncertainty, testing hypotheses, and modeling relationships. In this section, we connect the theory of quadratic forms to cornerstone sampling distributions, chi-square and D, and a foundational linear regression, showing how every step of inference relies on the properties we’ve already explored. [BKP24]

4.1 Quadratic Forms and the Chi-Square Distribution: Expressing the ”Sum of Squares”

For a random vector $X \in \mathbb{R}^n$ following a multivariate normal distribution $X \sim N(\mu, \Sigma)$, where $\mu \in \mathbb{R}^n$ is the mean vector and $\Sigma \in \mathbb{R}^{n \times n}$ is the symmetric, positive definite covariance matrix. In this case, quadratic forms provide a natural way to construct ”sum of squares” statistics, which are central to hypothesis testing.

4.1.1 The Simple Case: standard Normal Vectors

Recall the definition of quadratic form is $Q(x) = x^T A x$ for a symmetric matrix A . For a standard normal distribution vector $Z \sim N(0, I_n)$, the quadratic form $Q(Z) = Z^T I_n Z = \sum_{i=1}^n Z_i^2$ is just the sum of squared independent standard normal variables. By definition, this sum follows a chi-square distribution with n degrees of freedom, denoted $X^2(n) \sim Z^T I_n Z$

4.1.2 General Normal Vectors: Standardization via Quadratic Forms

For a non-standard normal vector $X \sim N(\mu, \Sigma)$, we can use the Spectral Theorem in Section 1.2 to "Standardize," or orthogonally diagonalize, X into a standard normal vector. Since Σ is the covariance matrix where $\Sigma_{ij} = \text{Cov}(X_i, X_j) = E[(X_i - \mu_i)(X_j - \mu_j)]$. By definition of covariance,

$$\text{Cov}(X_i, X_j) = E[(X_i - \mu_i)(X_j - \mu_j)] = E[(X_j - \mu_j)(X_i - \mu_i)] = \text{Cov}(X_j, X_i)$$

Lemma 4.1. *Covariance can be interpreted as an inner product when defined on a specific vector space of random variables. This allows us to use geometric intuition to understand statistical concepts.*

Proof. We can prove that Covariance is inner product on centered random variables first. Let us define an inner product $\langle X, Y \rangle = \text{Cov}(X, Y) = E[XY]$, since for centered variables X, Y , we have $E[X] = E[Y] = 0$.

1. Linearity:

$$\langle aX + bY, Z \rangle = E[(aX + bY)Z] = aE[XZ] + bE[YZ] = a\langle X, Z \rangle + b\langle Y, Z \rangle$$

2. Symmetry:

$$\langle X, Y \rangle = E[XY] = E[YX] = \langle Y, X \rangle$$

3. Positive Definiteness:

$$\forall X \in V, \langle X, X \rangle = \text{Cov}(X, X) = E[X^2] \geq 0; \langle X, X \rangle = 0 \iff E[X] = 0$$

Therefore, for centered variables, covariance is an inner product. Even though we can center most random variables by defining new distribution $Y = f(X)$ such that $E[Y] = 0$, but there are still some variables that cannot be centered. Then, covariance is not an inner product on these variables. $\square$

Thus, since Covariance is an inner product random variables that can be centered, for Normal Distributions, the Covariance is an inner product and symmetry and positive definiteness, allowing the Covariance Matrix Σ to be orthogonally diagonalizable by Spectral Theorem.

Let $\Sigma = PDP^T$, where $D = \text{diag}(\lambda_1, \dots, \lambda_n)$ is a diagonal matrix of positive eigenvalues. The inverse of Σ is also symmetric and positive definite: $\Sigma^{-1} = PD^{-1}P^T$.

We can define

$$Y = \Sigma^{-\frac{1}{2}}(X - \mu), \text{ such that } Y \sim B(0, I_n)$$

Thus we can express the quadratic form of X with Σ^{-1} [MP89]:

$$Q(X) = (X - \mu)^T \Sigma^{-1} (X - \mu) = (X - \mu)^T \Sigma^{-\frac{1}{2}} \Sigma^{-\frac{1}{2}} (X - \mu) = Y^T Y = \sum_{i=1}^n Y_i^2$$

Thus we proved $Q(X) \sim \chi^2(n)$, giving chi-square distribution a generalized linear algebra form. This result is foundational for the test of multivariable hypothesis, as the quadratic form here quantifies the distance between X and μ , and is scaled by the covariance matrix Σ .

4.2 Quadratic Forms in Linear Regression

Linear regression is one of the most widely used statistical methods, and at its core—residual sum of squares, regression sum of squares, and the F-test—are all quadratic forms.

Consider a linear regression model:

$$Y = X\beta + \epsilon$$

where $Y \in \mathbb{R}^n$ is the vector of observed responses, $X \in \mathbb{R}^{n \times p}$ is the design matrix (with p columns, including the column of 1s for the intercept), $\beta \in \mathbb{R}^p$ is the vector of unknown regression coefficients, $\epsilon \in \mathbb{R}^n$ is the vector of error, assumed to be $\epsilon \sim N(0, \sigma^2 I_n)$ (independent, normal errors).

4.2.1 Residual Sum of Squares (Sum of Squared Errors): A Quadratic Form

The fitted values of Y are $\bar{Y} = X\bar{\beta}$ where $\bar{\beta} = (X^T X)^{-1} X^T Y$ is the projection and least squares estimator. The residual sum of squares is the sum of squared residuals:

$$\text{Sum of Squared Errors} = \epsilon^2 = \epsilon^T \epsilon = (Y - X\hat{\beta})^T (Y - X\hat{\beta})$$

Let $H = X(X^T X)^{-1} X^T$, then $e = Y - HY = (I - H)Y$, we can rewrite SSE as $Y^T (I_n - H)Y$, which is a quadratic form.

4.2.2 Distribution of SSE: Chi-Square Distribution through Quadratic Forms

Under the regression model assumption, $Y \sim N(X\beta, \sigma^2 I_n)$. The quadratic form $\text{SSE} = Y^T M Y$ has a special distribution when scaled by σ^2 . The rank is $\text{tr}(M) = n - p$. Thus we can show that:

$$\frac{\text{SSE}}{\sigma^2} = \frac{Y^T M Y}{\sigma^2} \sim \chi^2(n - p)$$

where M is the projection matrix, and the residuals are orthogonal to the fitted values $MX = 0$.

4.2.3 Geometric Interpretation: Projections and Quadratic Forms

The projection matrix H and its complement $M = I_n - H$ are orthogonal projection matrices where H takes our observed response vector $\mathbf{Y}$ and projects it onto $\mathcal{C}(X)$ (the column space of the design matrix X), which is exactly the space of all possible fitted values $\hat{\mathbf{Y}} = X\hat{\beta}$. $M = I_n - H$ projects $\mathbf{Y}$ onto the orthogonal complement of $\mathcal{C}(X)$ (denoted $\mathcal{C}(X)^\perp$)—the set of all vectors that are perpendicular to every column of X , and where our residuals live.

What makes this projection framework so powerful is that projection matrices preserve squared lengths (a direct extension of the Pythagorean theorem to vector spaces). This means the residual sum of squares (SSE) isn't just a random sum of squared residuals—it's the squared length of the projection of $\mathbf{Y}$ onto $\mathcal{C}(X)^\perp$:

$$\text{SSE} = \|\mathbf{e}\|^2 = \|M\mathbf{Y}\|^2 = \mathbf{Y}^T M \mathbf{Y}.$$

4.3 Definition of the F-Distribution

A good way to enhance understanding is to anchor the F-distribution to quadratic forms (rather than just memorizing it) to keep the logic intact: if $U \sim \chi^2(d_1)$ and $V \sim \chi^2(d_2)$ are independent chi-square random variables (each of which is a quadratic form of a normal vector), then the ratio

$$F = \frac{U/d_1}{V/d_2}$$

follows an F-distribution with degrees of freedom (d_1, d_2) , denoted $F(d_1, d_2)$. The division by degrees of freedom is critical as it "normalizes" each chi-square term, so we're comparing ratios to ratios.

4.4 The F-Test in Regression: A Ratio of Quadratic Forms

When we want to test if a regression model has any predictive power (i.e., $H_0 : \beta_1 = \beta_2 = \dots = \beta_{p-1} = 0$, meaning all slope coefficients are zero), the F-test leverages quadratic forms at every step. Here's how we can break it down intuitively:

First, we split the total sum of squares (SST), which is the total variation in $\mathbf{Y}$, into two parts: variation explained by the regression (SSR) and variation left unexplained (SSE). SST itself is a quadratic form:

$$\text{SST} = \mathbf{Y}^T \left(I_n - \frac{1}{n} J_n \right) \mathbf{Y},$$

where J_n is the $n \times n$ matrix of all 1s (we can think of J_n as the "constant vector projector," since it maps $\mathbf{Y}$ onto the space of constant values). The decomposition is clean:

$$\text{SST} = \text{SSR} + \text{SSE}.$$

SSR (the "explained" variation) is also a quadratic form that isolates the variation captured by the regression (excluding the constant term):

$$\text{SSR} = \mathbf{Y}^T \left(H - \frac{1}{n} J_n \right) \mathbf{Y}.$$

Under the null hypothesis H_0 (no predictive power), two key things hold:

1. $\frac{\text{SSR}}{\sigma^2} \sim \chi^2(p-1)$ (the rank of $H - \frac{1}{n} J_n$ is $p-1$, so this quadratic form has $p-1$ degrees of freedom),
2. $\frac{\text{SSE}}{\sigma^2} \sim \chi^2(n-p)$ (from Section 2.3, our residual quadratic form),
3. Most importantly, SSR and SSE are independent—this is because the matrices $H - \frac{1}{n} J_n$ (for SSR) and $M = I_n - H$ (for SSE) are orthogonal (their product is zero, so their quadratic forms don't "overlap").

Putting this all together, the F-statistic for testing H_0 is just the ratio of these normalized quadratic forms:

$$F = \frac{\text{SSR}/(p-1)}{\text{SSE}/(n-p)} \sim F(p-1, n-p).$$

For me, this is the beauty of the quadratic form framework: the F-statistic isn't an arbitrary ratio, it's a natural consequence of decomposing variation in $\mathbf{Y}$ into orthogonal quadratic forms, each with known distributions. Every property of the F-test (its distribution, its independence assumptions, its degrees of freedom) traces back to the linear algebra of quadratic forms we explored earlier.

4.5 Summary: Quadratic Forms as a Unifying Language

In statistics, quadratic forms serve as a unifying language that connects linear algebra, geometry and inference. The chi-square distribution and F distribution are defined by quadratic forms of normal vectors. Linear regression also relies on quadratic forms to quantify the expression. What makes the connection so powerful is the properties of quadratic forms: symmetry, diagonalization, definiteness. They provide a shortcut to understanding statistical results. For example, the positive semidefiniteness of quadratic forms ensures SSE is non negative, and the expression of H ensures the fitted values are orthogonal to the residuals.

In short, quadratic forms are not just a "tool in statistics - they are the mathematical framework that makes classical inference possible. By understanding quadratic forms, we gain a deeper insight into why statistical tests work, how to interpret their results, and how to generalize them to more complex models.

5 Conclusion

Through the Spectral Theorem, quadratic forms admit a remarkably complete structural description: symmetry guarantees real eigenvalues, orthogonal diagonalization, and a clear geometric interpretation in terms of independent curvature along orthogonal directions. This framework not only simplifies classification, but also connects linear algebra directly to analysis and geometry, as seen through applications to optimization and the Hessian test. Quadratic forms therefore occupy a uniquely rigid and well-behaved position within multi-linear algebra.

The extension to cubic forms highlights the limits of this structure. While cubic forms arise naturally from determinants and norm-preserving bilinear multiplications, they resist the tools that make quadratic forms tractable: there is no symmetric matrix representation, no orthogonal diagonalization, and no analogue of the Spectral Theorem. Instead, their structure is governed by algebraic and multiplicative invariance rather than linear symmetry.

Conversely, the extension of quadratic forms to statistics underscores their exceptional utility in connecting disparate fields. As we saw in Section 4, quadratic forms serve as the mathematical backbone of classical inference, uniting linear algebra, geometry, and probability to enable hypothesis testing, regression analysis, and uncertainty quantification—applications that rely directly on the symmetric structure, diagonalization, and definiteness we've explored.

Together, these extensions—one revealing the limits of quadratic structure, the other showcasing its unmatched power—emphasize that the tractability and versatility of quadratic forms are exceptional, not typical, in the landscape of polynomial forms. This duality motivates the search for fundamentally different methods when studying higher-degree forms (like cubics) and using quadratic forms as a tool to other mathematical and statistical domains.

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